MUNIBA MANAAL

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with a BS in Artificial Intelligence and one year of industry experience turning AI into products that actually work. I build AI automation workflows, intelligent agents, and LLM-powered applications that take on real business tasks with little human effort. Skilled in Python, deep learning, computer vision, and LLM fine-tuning, and built an AI recruitment platform that cut resume screening time by 75%.

EDUCATION

COMSATS University Islamabad | Islamabad, Pakistan — Sep 2022 – Jun 2026

Bachelor of Science in Artificial Intelligence

Relevant Coursework: Machine Learning, Deep Learning, Artificial Neural Networks, Reinforcement Learning, Computer Vision, Speech Processing, Knowledge Representation & Reasoning, Natural Language Processing, Data Structures & Algorithms, Database Systems, Operating Systems, Web Technologies

PROFESSIONAL EXPERIENCE

Launchify LLC | Islamabad, Pakistan - *Hybrid* — September 2026 – Present

AI Engineer

- Design and build **AI automation workflows** and **intelligent agents** that handle business tasks end to end with minimal human input, such as outbound calling and meeting scheduling.
- Integrate LLMs and voice AI platforms (**Vapi**) with tools and APIs such as **Google Calendar** to deliver complete solutions.
- Developed AI products including voice calling agents and **facial emotion recognition** models.
- Fine-tuned the **Flan-T5** language model for text summarization and question-answering tasks.

YoungDevs | Remote

Python Developer Intern

- Built **REST APIs** in Python to support data exchange between application components and services.
- Cleaned, pre-processed, and transformed datasets using **Pandas** and **NumPy**, and performed detailed **exploratory data analysis (EDA)** to prepare model-ready data.
- Automated file handling and report generation tasks, reducing repetitive manual work.
- Implemented search algorithms and **object-oriented** solutions in Python for assigned development tasks.

PROJECTS

AURORA - AI-Powered Recruitment & Training Platform | Final Year Project — 2025 – 2026

- Built resume screening with **Sentence-BERT** and Two-Tower ranking (NDCG 0.705), cutting screening time by **75%**.
- Developed an AI interview agent (**Groq Llama**) with facial emotion detection using MobileNetV3 trained on AffectNet data (**92% accuracy**, 7 classes) and **MediaPipe** gaze tracking.
- Added personalized learning paths, course recommendations, and an HR analytics dashboard.

Dialogue Summarization - LLM Fine-Tuning | Generative AI, LLMs

- Fine-tuned **Flan-T5** on the DialogSum dataset with **full fine-tuning** and **PEFT/LoRA** (training only 1.4% of parameters), improving ROUGE-1 by 18.8 points over the base model; completed in DeepLearning.AI's Generative AI with LLMs course.

Sea Animal Image Classification | Deep Learning, Transfer Learning

- Fine-tuned an ImageNet pre-trained **ResNet50V2** in **TensorFlow/Keras** on a Kaggle sea animal image dataset for multi-class species classification.

Emotion Recognition from Radio Signals | Computer Vision & Deep Learning

- Fine-tuned a **DINOv2 Vision Transformer** on radio-signal emotion data for cross-modal emotion classification.

Breast Cancer Detection | Deep Learning, Healthcare AI

- Fine-tuned a pre-trained **ResNet50** in **TensorFlow** on breast ultrasound images (BUSI) to classify scans as normal, benign, or malignant, supporting early diagnosis.

IoT Intrusion Detection - Model Comparison | Machine Learning

- Compared **Decision Tree**, **Naive Bayes**, and **Neural Network** classifiers for binary and multiclass attack detection (DoS, Mirai, MITM ARP Spoofing, Scan) on the **IoTID20** network-traffic dataset, with data preprocessing, outlier removal, and confusion-matrix evaluation.

House Price Predictor | Machine Learning, Model Deployment

- Compared **Linear Regression**, **Decision Tree**, **SVM**, and **KNN** regression models, and deployed the best-performing model as a live web app on **Anvil**.

Connect 4 Game Agent | Game AI/ Game Theory, Python

- Built agents using **Monte Carlo Tree Search** and **Alpha-Beta Pruning**, benchmarked against minimax baselines.

Other Projects: Weather Forecasting (LSTM), Facial Emotion Recognition (CNN on FER-2013), Traffic Cone and Barricade Detection (Object Detection - YOLO), Movie Recommender (KNN), Music Generator (LSTM), Lyrics Generator (RNN), Travel & Budget Recommendation Expert System (Prolog), Text Classification with DL Architectures (FFN, CNN, RNN, Transformers), Google Playstore Apps Download Prediction Model, Pathfinding Game Agents (A*, Dijkstra), Genetic Algorithm Game, Obstacle-Avoiding

Robot (Arduino, ultrasonic sensors), Alif NGO System (Web App), Flight Management System, Amazon Clone E-Commerce Site (JavaScript), Atlas UI Design System, Scrollytelling Site (CSS scroll animations), Generative Art Playground (Canvas)

SKILLS

Generative AI & LLMs: LLM Fine-Tuning, LLM Evaluation & Benchmarking, Prompt Engineering, LangChain, Text Summarization, Question Answering
AI Agents & Automation: AI Agents, Workflow Automation, Game-Playing & Search Agents
Machine Learning: Supervised & Unsupervised Learning, Classification, Regression, Clustering, Dimensionality Reduction, Recommendation Systems
Deep Learning: Neural Networks (ANN, CNN, RNN, LSTM), Transformers, Transfer Learning, Time-Series Forecasting
Computer Vision: Image Classification, Object Detection & Tracking, Facial Emotion Recognition, OCR, Eye & Gaze Tracking
AI Foundations: Reinforcement Learning, Genetic Algorithms, Search & Optimization, Knowledge Representation & Reasoning, NLP, Speech Processing
Programming Languages: Python, JavaScript, C++, Java, Prolog
Frontend: HTML, CSS, JavaScript, React | **Backend:** Node.js, Django, FastAPI | **Tools:** Git, GitHub
Libraries & Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch, Hugging Face Transformers, LangChain, OpenCV, Matplotlib

CERTIFICATIONS

Machine Learning Specialization | DeepLearning.AI & Stanford
Deep Learning Specialization | DeepLearning.AI
LangChain for LLM Apps | DeepLearning.AI
LLM Benchmarking and Evaluation | Simplilearn
Generative AI with Large Language Models | DeepLearning.AI & AWS
IBM Full Stack Software Developer Certificate | IBM - In Progress
Flutter Bootcamp ’25 | GDGoC PIEAS
Google Soft Skills Program 2024 | PAFLA